

Periodic table

- an arrangement of elements in periods & groups and in order of increasing proton/atomic number.
- the left side of the periodic table is metals while the right side is non-metals.
- metals usually lose electrons to become stable
- non-metals usually gain electrons to become stable

groups/periods

- the group tells you how many outer shell electrons an element has which tells you how much it needs to become stable.
- the period tells you how many shells the element has
- * group IV elements have 4 electrons in outermost shell
- * period 2 elements have 2 shells
- * group I element will lose an electron to become stable thus having a +1 charge
- * group VII element will gain an electron to become stable thus having a -1 charge
- * elements in the same group have similar chemical properties because when atoms collide, their outermost electrons interact where elements of the same have the same number of outermost electrons.

periodic table can predict properties such as:

- boiling point
- melting point
- density
- reactivity

GROUP I

boiling/melting point :: decreases as you go down the group

density :: increases as you go down the group

reactivity :: increases as you go down the group

GROUP II

boiling/melting point :: N/A

density :: increases as you go down the group

reactivity :: increases as you go down the group

GROUP VII

boiling/melting point :: increases as you go down the group

density :: increases as you go down the group

reactivity :: decreases as you go down the group

GROUP VIII

boiling/melting point :: increases as you go down the group

density :: increases as you go down the group

reactivity :: increases as you go down the group

Reactivity series :

potassium, sodium, calcium, magnesium, aluminium, carbon, zinc, iron, hydrogen, copper, silver, gold

mnemonic

"Please send cats, monkeys and cute zebras into hot countries, signed Gordon"

GROUP VII :

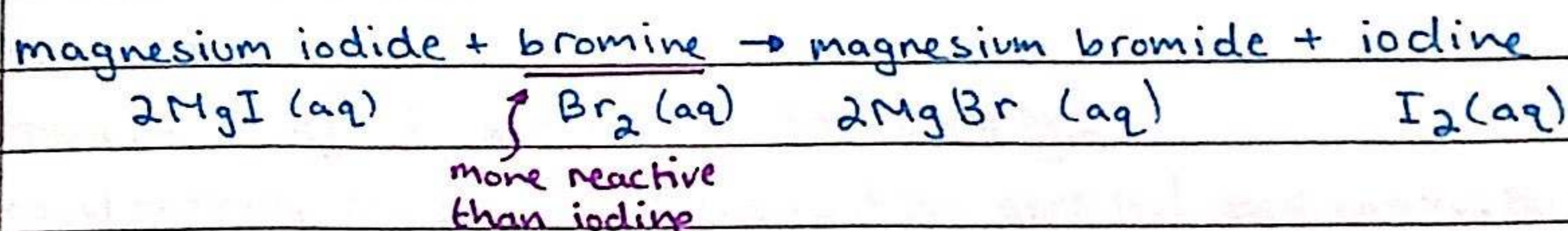
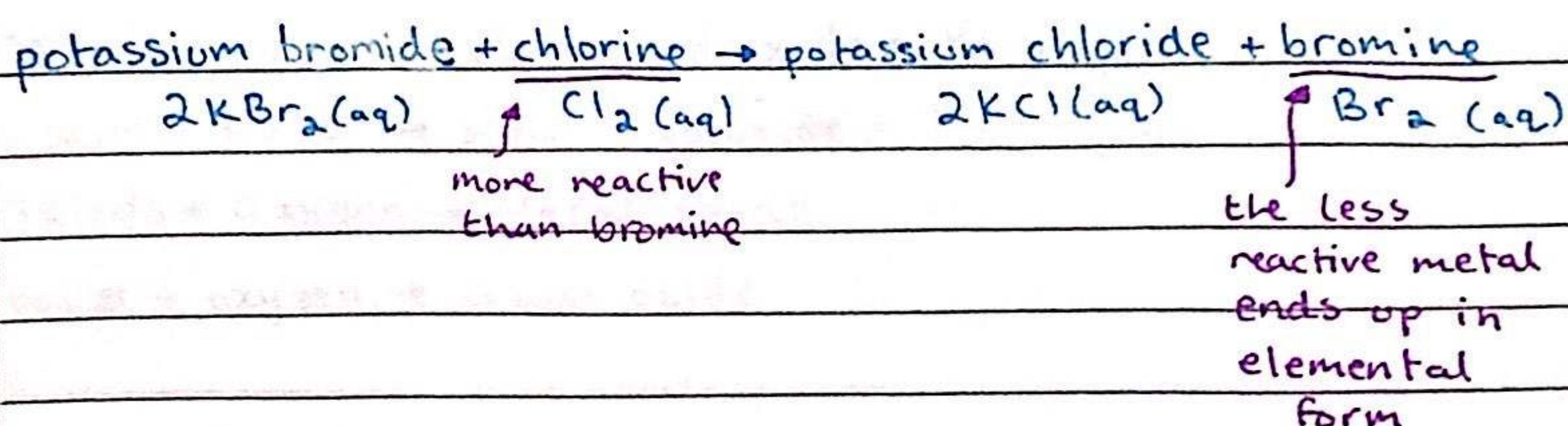
chlorine:- pale yellow-green gas

bromine:- red-brown liquid

iodine:- grey-black solid

displacement reactions

- a reaction where a more reactive element displaces or pushes out a less reactive element from a compound

TRANSITION METALS

- high densities
 - high melting points
 - form colored compounds
 - often act like catalysts
- they have ions with variable oxidation numbers
e.g iron(II) or iron(III)

GROUP VIII

- unreactive = outershell is already complete
- monatomic gases = they have high stability so they don't combine with other atoms.